

Qualification Pack



Assistant Technician - Street Light Installation & Maintenance

QP Code: PSS/Q2506

Version: 3.0

NSQF Level: 3

Power Sector Skill Council || Plot No. 4, Institutional Area, CBIP Building, Malcha Marg, Chanakyapuri
New Delhi-110021 || email: bishwajeet.singh@psscindia.org



Qualification Pack

Contents

PSS/Q2506: Assistant Technician - Street Light Installation & Maintenance	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
PSS/N2510: Perform installation of street lighting system	5
PSS/N2511: Perform testing and repairing of street lighting system	9
PSS/N1331: Apply basic health and safety practices for power related work	13
PSS/N1336: Working effectively with others	19
SGJ/N1702: Optimize resource utilization at workplace	24
DGT/VSQ/N0101: Employability Skills (30 Hours)	29
Assessment Guidelines and Weightage	34
<i>Assessment Guidelines</i>	34
<i>Assessment Weightage</i>	34
Acronyms	36
Glossary	37



Qualification Pack

PSS/Q2506: Assistant Technician - Street Light Installation & Maintenance

Brief Job Description

The incumbent in the job is responsible for installation, testing, commissioning, operations & maintenance of a streetlight network including poles, fixtures, cables, junction boxes, feeder pillars and other associated accessories.

Personal Attributes

The job requires the individual to work outdoors in all weather conditions and requires availability in the event of emergency situations. The job also requires the ability to lift large heavy objects and perform strenuous physical labor. The individual requires bending, climbing, pulling, carrying, kneeling, walking, and standing for significant periods of time.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N2510: Perform installation of street lighting system](#)
2. [PSS/N2511: Perform testing and repairing of street lighting system](#)
3. [PSS/N1331: Apply basic health and safety practices for power related work](#)
4. [PSS/N1336: Working effectively with others](#)
5. [SGJ/N1702: Optimize resource utilization at workplace](#)
6. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution, Operation and Maintenance - Distribution
Country	India
NSQF Level	3



Qualification Pack

Credits	10
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7413.9900
Minimum Educational Qualification & Experience	8th grade pass plus 2-year NTC plus 1 Year NAC (Electrician) OR 8th grade pass with 1 Year of experience Relevant Experience OR 5th grade pass with 4 Years of experience Relevant Experience
Minimum Level of Education for Training in School	5th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/03/2025
NSQC Approval Date	12/01/2023
Version	3.0
Reference code on NQR	2022/POW/PSSC/06922
NQR Version	2.0



Qualification Pack

PSS/N2510: Perform installation of street lighting system

Description

This OS unit is about performing the installation/erection of street light poles, fixtures, lighting system and laying underground wiring/cable in accordance with the company and Indian standards.

Scope

The scope covers the following :

- Street light installation
- Laying the underground cable

Elements and Performance Criteria

Street light installation

To be competent, the user/individual on the job must be able to:

- PC1.** carry out the root survey for planning of the street lighting system installation and cable lay work
- PC2.** identify tools and equipment required for street light installation
- PC3.** select appropriate type of lighting system based on the requirements and parameters i.e. wattage and lumen level
- PC4.** test light fixture and drivers, prior to the installation
- PC5.** install protection devices such as surge protection device, voltage fluctuation, over voltage protection etc.
- PC6.** connect the streetlight fixture with the power line meant for street lighting
- PC7.** set timing on lighting system control switch for automatic switch on and off operation of light

Laying the underground cable

To be competent, the user/individual on the job must be able to:

- PC8.** select and lay underground cable for street lighting as per the standards
- PC9.** connect power supply from distribution transformer to the street lighting system
- PC10.** troubleshoot and rectify all the problems involved in underground electrical wiring
- PC11.** report to supervisor or engineer any problem if needed in laying down the wiring
- PC12.** perform primary testing on the cables and connections of street light

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards, policies and installation procedures followed in the organization
- KU2.** common electricity terminology e.g. current, voltage, resistance, kilowatt (kw), kilowatt hour(kwh)



Qualification Pack

- KU3.** types of street light fixtures such as sodium vapour lamp, CFL, Halogen, LED, Mercury lamp etc. and their selection parameters such as wattage and lumen level of street light
- KU4.** various type of protection driver/devices -surge protection, voltage fluctuation, over voltage etc.
- KU5.** the wiring and layout diagrams of street lights
- KU6.** the cabling system for street light
- KU7.** use of control switch and timer for automatic switch on and switch off operation
- KU8.** use of lux meter, multimeter and other testing equipment
- KU9.** ratings and specifications of cables, fuses, switches and wires
- KU10.** Standard Operating Procedures recommended by the manufacturer for using machines, tools and equipment

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write reports and observations related to work in English/regional language
- GS2.** attentively listen and comprehend the information given by the supervisor/team members
- GS3.** read and interpret the equipments operational/technical documents
- GS4.** discuss task lists and job requirements with supervisor and co-workers
- GS5.** follow organization rule-based decision making process in consultation of the supervisor
- GS6.** analyse the complexity of work to determine if it can be successfully carried out or needs to be referred to a superior/specialist
- GS7.** recognise a workplace problem and take suitable action to resolve it

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Street light installation</i>	13	48	-	-
PC1. carry out the root survey for planning of the street lighting system installation and cable lay work	1	5	-	-
PC2. identify tools and equipment required for street light installation	2	6	-	-
PC3. select appropriate type of lighting system based on the requirements and parameters i.e. wattage and lumen level	2	6	-	-
PC4. test light fixture and drivers, prior to the installation	2	8	-	-
PC5. install protection devices such as surge protection device, voltage fluctuation, over voltage protection etc.	2	9	-	-
PC6. connect the streetlight fixture with the power line meant for street lighting	2	9	-	-
PC7. set timing on lighting system control switch for automatic switch on and off operation of light	2	5	-	-
<i>Laying the underground cable</i>	9	30	-	-
PC8. select and lay underground cable for street lighting as per the standards	2	6	-	-
PC9. connect power supply from distribution transformer to the street lighting system	1	6	-	-
PC10. troubleshoot and rectify all the problems involved in underground electrical wiring	2	6	-	-
PC11. report to supervisor or engineer any problem if needed in laying down the wiring	2	6	-	-
PC12. perform primary testing on the cables and connections of street light	2	6	-	-
NOS Total	22	78	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2510
NOS Name	Perform installation of street lighting system
Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution, Operation and Maintenance - Distribution
NSQF Level	3
Credits	3.5
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/03/2025
NSQC Clearance Date	12/01/2023



Qualification Pack

PSS/N2511: Perform testing and repairing of street lighting system

Description

This OS unit is about performing testing and repairing activities on the street light system

Scope

The scope covers the following :

- Inspect and test the street lights
- Repair and replace the light components

Elements and Performance Criteria

Inspect and test the street lights

To be competent, the user/individual on the job must be able to:

- PC1.** identify testing equipment such as tester, multimeter, wire, bulb etc. required during maintenance work
- PC2.** check all the intersections, joints, junction box, street light fixture, light bulb etc. for any defect as per the checklist
- PC3.** use lux meter to check the lux level of light fixture
- PC4.** test the supply across the MCB / ON-OFF switch and all the joints and intersections in the wiring
- PC5.** test the cable and lamp head by multimeter or tester or clipon meter for checking the continuity of supply
- PC6.** check the fixture and other parts of street light system

Repair and replace the light components

To be competent, the user/individual on the job must be able to:

- PC7.** check the photoelectric control relay and surge protection devices of lighting system and repair or replace in case of any defect or problem as per the organizational procedures
- PC8.** troubleshoot underground electrical wiring and repair it in case of any problem
- PC9.** check the street light equipment including drivers, heads, poles, controllers, lights, circuitry, switches, fuses, and cabinet and repair or replace in case of any defect or problem as per the organizational procedures
- PC10.** replace and retrofit the existing light fixture with modern light fixture, where required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant street lighting operational and maintenance standards followed in the organization
- KU2.** technical specification of lights, associated components and protection devices
- KU3.** the wiring diagram and cabling system of the street light
- KU4.** use of lux meter, multimeter and other testing equipment



Qualification Pack

KU5. various testing, inspection methods and repairing procedure of street lighting system

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write reports and observations related to work in English/regional language
- GS2.** attentively listen and comprehend the information given by the supervisor/team members
- GS3.** read and interpret the testing and maintenance documents
- GS4.** discuss task lists and job requirements with supervisor and co-workers
- GS5.** analyze the complexity of work to determine if it can be successfully carried out or needs to be referred to a superior/specialist
- GS6.** recognize a workplace problem and take suitable action to resolve it

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspect and test the street lights</i>	16	46	-	-
PC1. identify testing equipment such as tester, multimeter, wire, bulb etc. required during maintenance work	3	10	-	-
PC2. check all the intersections, joints, junction box, street light fixture, light bulb etc. for any defect as per the checklist	3	10	-	-
PC3. use lux meter to check the lux level of light fixture	2	6	-	-
PC4. test the supply across the MCB / ON-OFF switch and all the joints and intersections in the wiring	2	6	-	-
PC5. test the cable and lamp head by multimeter or tester or clipon meter for checking the continuity of supply	3	7	-	-
PC6. check the fixture and other parts of street light system	3	7	-	-
<i>Repair and replace the light components</i>	10	28	-	-
PC7. check the photoelectric control relay and surge protection devices of lighting system and repair or replace in case of any defect or problem as per the organizational procedures	2	7	-	-
PC8. troubleshoot underground electrical wiring and repair it in case of any problem	2	7	-	-
PC9. check the street light equipment including drivers, heads, poles, controllers, lights, circuitry, switches, fuses, and cabinet and repair or replace in case of any defect or problem as per the organizational procedures	4	9	-	-
PC10. replace and retrofit the existing light fixture with modern light fixture, where required	2	5	-	-
NOS Total	26	74	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2511
NOS Name	Perform testing and repairing of street lighting system
Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution, Operation and Maintenance - Distribution
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/03/2025
NSQC Clearance Date	12/01/2023



Qualification Pack

PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, organizational assets and the environment.

Scope

The scope covers the following :

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)
- PC3.** use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment
- PC4.** follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)
- PC5.** ensure positive isolation of electrical equipment and system as per the given standards
- PC6.** identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.
- PC7.** use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)
- PC8.** use proper scaffolds, elevated platforms and ladders for working at height
- PC9.** inspect various tools and plants (T & P) routinely for any signs of oil, water leakages
- PC10.** store flammable materials and machine lubricating oil safely and correctly



Qualification Pack

PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)

PC14. use appropriate rescue techniques during fire hazard

PC15. use good housekeeping practices at all times to avoid accidental fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)

PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real

PC18. write an incident report and send the same to the authorized person

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. names (and job titles if applicable) and location of all documents along with the people responsible for health and safety in a workplace

KU2. possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity (such as drunkenness), health hazards (such as untreated injuries and contagious illness), not taking safety precautions

KU3. safe working practices when working with tools, machines

KU4. positive isolation of electrical equipment and system

KU5. use of emission and pollution control devices and measures taken to control pollution

KU6. preventative and remedial actions to be taken in the case of exposure to toxic materials solvents, flux, lead etc.

KU7. precautionary measures taken to prevent fire accident

KU8. various causes of fire (Causes of fires: heating of metal, spontaneous ignition, sparking, electrical heating, loose fires (smoking, welding, etc.), chemical fires, etc.)

KU9. different materials (sand, water, foam, CO₂, dry powder) used in fire extinguisher and proper method of using it to extinguish fire

KU10. various types of safety signs with their names

KU11. basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries basic techniques of bandaging

KU12. artificial respiration and the CPR Process



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read & write an accident/ incident report in local language and/or English
- GS2.** read and comprehend basic content to read labels, charts, signages, manuals etc.
- GS3.** question co-workers appropriately and give clear instruction to them and sub ordinates in order to clarify instructions
- GS4.** remain congenial while discussing and debating issues with co-workers
- GS5.** observe appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS6.** plan and organize work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS7.** build customer relationships and use customer centric approach
- GS8.** seek assistance from other sources to resolve problem and report problems that cannot be resolved to the appropriate authority

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	25	36	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)	3	5	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)	3	2	-	-
PC3. use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment	3	5	-	-
PC4. follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)	2	2	-	-
PC5. ensure positive isolation of electrical equipment and system as per the given standards	2	4	-	-
PC6. identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.	2	4	-	-
PC7. use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)	2	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. use proper scaffolds, elevated platforms and ladders for working at height	2	3	-	-
PC9. inspect various tools and plants (T & P) routinely for any signs of oil, water leakages	2	3	-	-
PC10. store flammable materials and machine lubricating oil safely and correctly	2	2	-	-
PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	3	-	-
PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	3	-	-
<i>Follow fire safety practices</i>	6	12	-	-
PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)	3	5	-	-
PC14. use appropriate rescue techniques during fire hazard	2	5	-	-
PC15. use good housekeeping practices at all times to avoid accidental fire	1	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	13	-	-
PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)	3	6	-	-
PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real	2	5	-	-
PC18. write an incident report and send the same to the authorized person	3	2	-	-
NOS Total	39	61	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1.5
Version	2.0
Next Review Date	31/08/2026



Qualification Pack

PSS/N1336: Working effectively with others

Description

This unit covers basic practices that improve effectiveness of working with others in an organizational set-up

Scope

The scope covers the following :

- Work effectively with others
- Respect diversity

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** accept information and instructions from the supervisor and fellow workers and seek clarification when required
- PC2.** pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt
- PC3.** help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency
- PC4.** use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism
- PC5.** listen attentively and respond appropriately while interacting with others at work
- PC6.** act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)
- PC7.** escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict

Respect diversity

To be competent, the user/individual on the job must be able to:

- PC8.** respect gender diversity in the workplace
- PC9.** use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive
- PC10.** communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture
- PC11.** recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same

Knowledge and Understanding (KU)



Qualification Pack

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** key elements of active listening and assertive communication
- KU6.** importance of teamwork in organizational and individual success
- KU7.** ethics and disciplined behavior for professional success
- KU8.** expressing and addressing grievances appropriately and managing interpersonal conflict effectively
- KU9.** harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
- KU10.** organizational policy/ies for harassment and discrimination

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
- GS2.** read basic terms and terminologies to accurately interpret work related documents, labels, supervisor instructions in the local language
- GS3.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans and other associated requirements
- GS4.** display active listening skills while interacting with co-workers and other in the workplace
- GS5.** follow organization's rule-based decision-making process
- GS6.** use appropriate planning to maintain a smooth relationship with fellow team members
- GS7.** meet customer requirements
- GS8.** critically evaluate operation parameters in relation to system normality

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with others</i>	18	32	-	-
PC1. accept information and instructions from the supervisor and fellow workers and seek clarification when required	2	4	-	-
PC2. pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt	1	4	-	-
PC3. help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency	2	4	-	-
PC4. use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism	4	6	-	-
PC5. listen attentively and respond appropriately while interacting with others at work	3	6	-	-
PC6. act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)	3	5	-	-
PC7. escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict	3	3	-	-
<i>Respect diversity</i>	16	24	-	-
PC8. respect gender diversity in the workplace	4	6	-	-
PC9. use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive	4	6	-	-
PC10. communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	4	6	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same	4	6	-	-
NOS Total	34	56	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	2.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

SGJ/N1702: Optimize resource utilization at workplace

Description

This unit is about adopting sustainable practices and optimizing use of resources, especially material, energy and waste, in day-to-day operations at work

Scope

The scope covers the following :

- Material conservation practices
- Energy/electricity conservation practices
- Effective waste management/recycling practices

Elements and Performance Criteria

Material conservation practices

To be competent, the user/individual on the job must be able to:

- PC1.** identify ways to optimize usage of material including water in various tasks/activities/processes
- PC2.** check for spills/leakages in various tasks/activities/processes
- PC3.** plug spills/leakages and escalate to appropriate authority if unable to rectify
- PC4.** carry out routine cleaning of tools, machines and equipment

Energy/electricity conservation practices

To be competent, the user/individual on the job must be able to:

- PC5.** identify ways to optimize usage of electricity/energy in various tasks/activities/processes
- PC6.** check if the equipment/machine is functioning normally before commencing work and rectify wherever required
- PC7.** report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment
- PC8.** ensure electrical equipment and appliances are properly connected and turned off when not in use

Effective waste management/recycling practices

To be competent, the user/individual on the job must be able to:

- PC9.** identify recyclable and non-recyclable, and hazardous waste generated
- PC10.** segregate waste into different categories
- PC11.** dispose non-recyclable waste appropriately
- PC12.** deposit recyclable and reusable material at identified location
- PC13.** follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



Qualification Pack

- KU1.** potential hazards, risks and threats based on the nature of work
- KU2.** layout of the workstation and electrical and thermal equipment used
- KU3.** organizations procedures for minimizing waste
- KU4.** efficient and inefficient utilization of material and water
- KU5.** ways of efficiently managing material and water in the process
- KU6.** basics of electricity and prevalent energy efficient devices
- KU7.** ways to recognize common electrical problems
- KU8.** common practices of conserving electricity
- KU9.** usage of different colours of dustbins
- KU10.** categorization of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU11.** waste management and methods of waste disposal
- KU12.** common sources of pollution and ways to minimize it

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data on waste disposal at workplace
- GS2.** complete statutory documents relevant to safety and hygiene
- GS3.** read Standard Operating Practices (SOP) documents
- GS4.** communicate with colleagues on the significance of greening of jobs
- GS5.** make timely decisions for efficient utilization of resources
- GS6.** complete tasks efficiently and accurately within stipulated time
- GS7.** work with supervisors/team members to carry out work related tasks
- GS8.** identify cause and effect of greening of jobs

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Material conservation practices</i>	4	8	-	-
PC1. identify ways to optimize usage of material including water in various tasks/activities/processes	1	2	-	-
PC2. check for spills/leakages in various tasks/activities/processes	1	2	-	-
PC3. plug spills/leakages and escalate to appropriate authority if unable to rectify	1	2	-	-
PC4. carry out routine cleaning of tools, machines and equipment	1	2	-	-
<i>Energy/electricity conservation practices</i>	4	8	-	-
PC5. identify ways to optimize usage of electricity/energy in various tasks/activities/processes	1	2	-	-
PC6. check if the equipment/machine is functioning normally before commencing work and rectify wherever required	1	2	-	-
PC7. report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment	1	2	-	-
PC8. ensure electrical equipment and appliances are properly connected and turned off when not in use	1	2	-	-
<i>Effective waste management/recycling practices</i>	5	10	-	-
PC9. identify recyclable and non-recyclable, and hazardous waste generated	1	2	-	-
PC10. segregate waste into different categories	1	2	-	-
PC11. dispose non-recyclable waste appropriately	1	2	-	-
PC12. deposit recyclable and reusable material at identified location	1	2	-	-
PC13. follow processes specified for disposal of hazardous waste	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	13	26	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1702
NOS Name	Optimize resource utilization at workplace
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Resource Optimization
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	25/11/2021
Next Review Date	19/06/2025
NSQF Clearance Date	25/11/2021



Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team



Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services



Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2026
Next Review Date	18/02/2029
NSQC Clearance Date	18/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment Guidelines will be same as that of QP

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N2510.Perform installation of street lighting system	22	78	-	-	100	40
PSS/N2511.Perform testing and repairing of street lighting system	26	74	-	-	100	20
PSS/N1331.Apply basic health and safety practices for power related work	39	61	0	0	100	10
PSS/N1336.Working effectively with others	34	56	0	0	90	10
SGJ/N1702.Optimize resource utilization at workplace	13	26	-	-	39	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	154	325	0	0	479	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.